

```
[19]: import pandas as pd
      a=pd.Series([1,2,3,4,6])
      print(a)
      print(a.values)
      print(a.index)
      print(a.dtype)

      b=pd.Series([1.3,2.6,3.7,4.8,5.9])
      print(b)
      print(b.values)
      print(b.index)
      print(b.dtype)

      c=pd.Series([True,False,True])
      print(c)
      print(c.values)
      print(c.index)
      print(c.dtype)

      # all types
      d=pd.Series([True,6.2,])
      print(d)
      print(d.values)
      print(d.index)
      print(d.dtype)

      e=pd.Series(["harsh","rithu","rithi","priya","renu","divya"])
      print(e)
      print(e.values)
      print(e.index)
      print(e.dtype)
      # accessing elements
      print(e[1])
```

```
e=pd.Series(["harsh","rithu","rithi","priya","renu","divya"])
print(e)
print(e.values)
print(e.index)
print(e.dtype)
# accessing elements
print(e[1])
#access elements using loc
print(e.loc[1])
#slicing
print(e[1:3])
print(e.loc[1:3]) # it access the stop with index value
print(e.iloc[2:5]) # it only shows the val before stop val
```

```
0    1
1    2
2    3
3    4
4    6
dtype: int64
[1 2 3 4 6]
RangeIndex(start=0, stop=5, step=1)
int64
0    1.3
1    2.6
2    3.7
3    4.8
4    5.9
dtype: float64
[1.3 2.6 3.7 4.8 5.9]
RangeIndex(start=0, stop=5, step=1)
float64
0    True
1   False
2    True
```

```

float64
0      True
1     False
2      True
dtype: bool
[ True False  True]
RangeIndex(start=0, stop=3, step=1)
bool
0      True
1      6.2
dtype: object
[True 6.2]
RangeIndex(start=0, stop=2, step=1)
object
0     harsh
1     rithu
2     rithi
3     priya
4     renu
5     divya
dtype: object
['harsh' 'rithu' 'rithi' 'priya' 'renu' 'divya']
RangeIndex(start=0, stop=6, step=1)
object
rithu
rithu
1     rithu
2     rithi
dtype: object
1     rithu
2     rithi
3     priya
dtype: object
2     rithi
3     priya
4     renu
dtype: object

```

```
[49]: import pandas as pd
arr=pd.Series([1,2,3,4,56],index=['A','B','C','D','E'])
print(arr[['A','C','E']])
print(arr)
print(arr['E']) # using label we can access
#vectorized operation-no loop required
print(arr+5)
print(arr.iloc[1]) # iloc access based on position
print(arr.loc['A'])
print(arr.sum())
print(arr.std())
print(arr.max())
print(arr.min())
print(arr.mean())
print(arr.median())
#boolean indexing
print(arr[arr>30])
#count values
print(len(arr))
#it return individual value count
print(arr.value_counts())
#string operation
name=pd.Series(['jack','vishnu','anitha','sashmitha','srinivas'])
print(name.str.upper())
print(name.str.len())
```

```
A      1
C      3
E     56
dtype: int64
A      1
B      2
C      3
D      4
E     56
```

```
A      1
C      3
E     56
dtype: int64
A      1
B      2
C      3
D      4
E     56
dtype: int64
56
A      6
B      7
C      8
D      9
E     61
dtype: int64
2
1
66
23.95203540411545
56
1
13.2
3.0
E     56
dtype: int64
5
1      1
2      1
3      1
4      1
56     1
Name: count, dtype: int64
0      JACK
1     VISHNU
2     ANITHA
```

```
Name: count, dtype: int64
```

```
0      JACK
```

```
1     VISHNU
```

```
2     ANITHA
```

```
3  SASHMITHA
```

```
4   SRINIVAS
```

```
dtype: object
```

```
0      4
```

```
1      6
```

```
2      6
```

```
3      9
```

```
4      8
```

```
dtype: int64
```

```
[41]: import pandas as pd
a=pd.Series({
    "gayathri":16,
    "rithanyaa":38,
    "bavanthika":8})
print(a)
print(["rithanyaa"])
```

```
gayathri      16
```

```
rithanyaa     38
```

```
bavanthika     8
```

```
dtype: int64
```

```
['rithanyaa']
```

```
[46]: #index alignment
s1=pd.Series([10,20,30],['A','B','C'])
s2=pd.Series([1,2,3],['B','C','D'])
print(s1+s2)
s=pd.Series([1,2,3,4,5])
print(s.count())
print(s.unique())
print(s.nunique()) # it show the count of unique values
```

```
A    NaN
B    21.0
C    32.0
D    NaN
dtype: float64
5
[1 2 3 4 5]
5
```

```
[120]: import pandas as pd
#method1 : from dictionary
data={
    'name':['monika','nagul','prem'],
    'rollno':[27,20,24],
    'Age':[19,20,21]}
df=pd.DataFrame(data)
print(df)
#from series to dataframe
s1=pd.Series([1,2,3,4,5])
s2=pd.Series([5.6,9.54,9.4,54.6,6])
s3=pd.Series([True,False,True,False])
df=pd.DataFrame({'sno':s1,
                 'cgpa':s2,
                 'attendance':s3})

print(df)
data=[
    ['athish', 22,98],
    ['rithikaa', 20,89],
    ['prathiksha',20,97]]
df=pd.DataFrame(data,columns=["name","Age","Mark"]) # here it will dispaly as matrix
print(df)
#list of dictionary
data=[
    {"name":"sakthi","course":"data science","grade":'A'},
    {"name":"madhu","course":"AIIML","grade":'A+'},
```

```
#list of dictionary
data=[
    {"name":"sakthi","course":"data science","grade":'A'},
    {"name":"madhu","course":"AIIML","grade":'A+'},
    {"name":"ragavi","course":"AIIML","grade":'A++'}]
print(df)

df=pd.DataFrame(data,index=["std1","std2","std3"]) # without index run and see
#adding one extra columns
df['marks']=[98,87,67]
print(df)

#adding one extra row
std4=pd.DataFrame([{'name':'mahi','course':'DS','grade':'O','marks':78}],index=['std4'])
df=pd.concat([df,std4])
print(df)
#creating multiple rows
students=pd.DataFrame([{'name':'sanjeev','course':'CSDA','grade':'O+','marks':98},
                        {'name':'divya','course':'AIIML','grade':'O++','marks':98}],index=['std5','std6'])
df=pd.concat([df,students])
print(df)

df['marks']=[98,87,67,78,98,98]
print(df['marks'])

#column access by bracket notation and dot notation
print(df.course)
print(df[['grade','marks']])

#shape attribute
print(df.shape)

#find no of row
print(df.shape[0])
```

```

#find no of columns
print(df.shape[1])

#Dtype attribute
#used to check data type of all columns
print(df.dtypes)

#info
print(df.info())
print(df.describe()) # it will print stastical value

#select specific row
print(df.loc[['std1','std2']])
print(df.iloc[3])

#update value
df.loc['std3','grade']='B++'
print(df)

#delete column
df.drop('grade', axis = 1,inplace=True) #inplace means changes made in the original frame directly
print(df)

#delete row
df.drop('std6',axis=0,inplace=False)
print(df)
df=df.drop('std6',axis=0,)
print(df)

#sorting
df.sort_values('marks',inplace=True) # asc default
print(df)

#df.sort_values('marks',inplace=False,inplace=True)
    
```

```
df.sort_values('marks',ascending=False,inplace=True)
print(df)
```

```
#stastical function for particular column
print(df['marks'].max())
```

	name	rollno	Age
0	monika	27	19
1	nagul	20	20
2	prem	24	21

	sno	cgpa	attendance
0	1	5.60	True
1	2	9.54	False
2	3	9.40	True
3	4	54.60	False
4	5	6.00	NaN

	name	Age	Mark
0	athish	22	98
1	rithikaa	20	89
2	prathiksha	20	97

	name	Age	Mark
0	athish	22	98
1	rithikaa	20	89
2	prathiksha	20	97

	name	course	grade	marks
std1	sakthi	data science	A	98
std2	madhu	AIML	A+	87
std3	ragavi	AIML	A++	67

	name	course	grade	marks
std1	sakthi	data science	A	98
std2	madhu	AIML	A+	87
std3	ragavi	AIML	A++	67
std4	mahi	DS	O	78

	name	course	grade	marks
std1	sakthi	data science	A	98
std2	madhu	AIML	A+	87

	name	course	grade	marks
std1	sakthi	data science	A	98
std2	madhu	AIML	A+	87
std3	ragavi	AIML	A++	67
std4	mahi	DS	O	78
std5	sanjeev	CSDA	O+	98
std6	divya	AIML	O++	98

std1 98

std2 87

std3 67

std4 78

std5 98

std6 98

Name: marks, dtype: int64

std1 data science

std2 AIML

std3 AIML

std4 DS

std5 CSDA

std6 AIML

Name: course, dtype: object

	grade	marks
std1	A	98
std2	A+	87
std3	A++	67
std4	O	78
std5	O+	98
std6	O++	98

(6, 4)

6

4

name object

course object

grade object

marks int64

dtype: object

(class 'pandas.core.frame.DataFrame')

```
<class 'pandas.core.frame.DataFrame'>
```

```
Index: 6 entries, std1 to std6
```

```
Data columns (total 4 columns):
```

#	Column	Non-Null Count	Dtype
0	name	6 non-null	object
1	course	6 non-null	object
2	grade	6 non-null	object
3	marks	6 non-null	int64

```
dtypes: int64(1), object(3)
```

```
memory usage: 240.0+ bytes
```

```
None
```

	marks
count	6.000000
mean	87.666667
std	12.971764
min	67.000000
25%	80.250000
50%	92.500000
75%	98.000000
max	98.000000

	name	course	grade	marks
std1	sakthi	data science	A	98
std2	madhu	AIML	A+	87
name	mahi			
course	DS			
grade	0			
marks	78			

```
Name: std4, dtype: object
```

	name	course	grade	marks
std1	sakthi	data science	A	98
std2	madhu	AIML	A+	87
std3	ragavi	AIML	B++	67
std4	mahi	DS	0	78
std5	sanjeev	CSDA	0+	98
std6	divya	AIML	0++	98

```
std4      mahi      DS      0      78
std5      sanjeev    CSDA    0+     98
std6      divya      AIML    0++    98
name      course  marks
std1      sakthi    data science 98
std2      madhu     AIML      87
std3      ragavi    AIML      67
std4      mahi      DS        78
std5      sanjeev    CSDA      98
std6      divya      AIML      98
name      course  marks
std1      sakthi    data science 98
std2      madhu     AIML      87
std3      ragavi    AIML      67
std4      mahi      DS        78
std5      sanjeev    CSDA      98
std6      divya      AIML      98
name      course  marks
std1      sakthi    data science 98
std2      madhu     AIML      87
std3      ragavi    AIML      67
std4      mahi      DS        78
std5      sanjeev    CSDA      98
name      course  marks
std3      ragavi    AIML      67
std4      mahi      DS        78
std2      madhu     AIML      87
std1      sakthi    data science 98
std5      sanjeev    CSDA      98
name      course  marks
std1      sakthi    data science 98
std5      sanjeev    CSDA      98
std2      madhu     AIML      87
std4      mahi      DS        78
std3      ragavi    AIML      67
98
```

```
[132]: import pandas as pd
#read csv files # usecols if u need specific column
df=pd.read_csv('customers-100.csv',usecols=["Last Name","Country"])
print(df)
#read only 1st 5 data
print(df.head())

#read only last 5 data
print(df.tail())
print(df.to_string())
print(df['Country'])
```

	Last Name	Country
0	Baxter	Chile
1	Lozano	Djibouti
2	Berry	Antigua and Barbuda
3	Olsen	Dominican Republic
4	Bender	Slovakia (Slovak Republic)
..	...	...
95	Greer	Guyana
96	Atkinson	Sri Lanka
97	Guerra	Solomon Islands
98	Farmer	Aruba
99	Haynes	Honduras

[100 rows x 2 columns]

	Last Name	Country
0	Baxter	Chile
1	Lozano	Djibouti
2	Berry	Antigua and Barbuda
3	Olsen	Dominican Republic
4	Bender	Slovakia (Slovak Republic)

	Last Name	Country
95	Greer	Guyana
96	Atkinson	Sri Lanka

	Last Name	Country
95	Greer	Guyana
96	Atkinson	Sri Lanka
97	Guerra	Solomon Islands
98	Farmer	Aruba
99	Haynes	Honduras
	Last Name	Country
0	Baxter	Chile
1	Lozano	Djibouti
2	Berry	Antigua and Barbuda
3	Olsen	Dominican Republic
4	Bender	Slovakia (Slovak Republic)
5	Downs	Bosnia and Herzegovina
6	Peck	Pitcairn Islands
7	Mullen	Bulgaria
8	Meyers	Cyprus
9	Gallagher	Timor-Leste
10	Schroeder	Guernsey
11	Dodson	Vietnam
12	Mata	Togo
13	Cox	Sri Lanka
14	Lutz	Singapore
15	Beasley	Oman
16	Foley	Western Sahara
17	Mata	Mozambique
18	Jacobson	South Georgia and the South Sandwich Islands
19	Kirk	French Polynesia
20	Frye	Malta
21	Houston	Netherlands
22	Howard	Paraguay
23	Valenzuela	Lao People's Democratic Republic
24	Wilcox	Albania
25	Moody	Panama
26	Poole	Belarus
27	Harper	Switzerland
28	Fitzpatrick	Saint Vincent and the Grenadines
29	Bradshaw	Tanzania

30	Tate	Zimbabwe
31	Trujillo	Denmark
32	Mejia	Liechtenstein
33	Santana	United States of America
34	Moon	Bahamas
35	Haley	Belize
36	Sosa	Uruguay
37	Gay	Solomon Islands
38	Esparza	Montenegro
39	Garza	Poland
40	Gentry	Dominican Republic
41	Dunn	Burkina Faso
42	Simpson	Bolivia
43	Greer	Bulgaria
44	Frost	Monaco
45	Montes	Palau
46	Guzman	Uzbekistan
47	Mccall	Ghana
48	Cohen	United Arab Emirates
49	Mccann	Belarus
50	Hawkins	Canada
51	Crawford	Algeria
52	Goodwin	Swaziland
53	Richard	Madagascar
54	West	Ecuador
55	Haas	Palestinian Territory
56	Maldonado	Saint Barthelemy
57	Parrish	United Arab Emirates
58	Hill	South Georgia and the South Sandwich Islands
59	Hardy	Sao Tome and Principe
60	Pham	Portugal
61	Harris	Togo
62	Jimenez	Ethiopia
63	Hutchinson	Netherlands
64	Lynch	Liberia
65	Lambert	Lithuania
66	Cannon	New Caledonia

66	Cannon	New Caledonia
67	Alvarado	Kiribati
68	Dudley	French Southern Territories
69	Good	Canada
70	Buck	Iran
71	Roth	Morocco
72	Keller	Zimbabwe
73	Benson	Russian Federation
74	Stein	Solomon Islands
75	Riddle	United States Virgin Islands
76	Zuniga	Kyrgyz Republic
77	Mcmahon	Myanmar
78	Pennington	Dominica
79	Blair	Iceland
80	Jennings	Micronesia
81	Mckay	Saint Vincent and the Grenadines
82	Moreno	Isle of Man
83	Davidson	Oman
84	Holt	Fiji
85	Cunningham	New Zealand
86	Woods	Maldives
87	Vargas	Benin
88	Kane	Yemen
89	Flowers	Namibia
90	Chavez	Bhutan
91	Foley	Hungary
92	Ayers	Anguilla
93	Young	Solomon Islands
94	Douglas	Mali
95	Greer	Guyana
96	Atkinson	Sri Lanka
97	Guerra	Solomon Islands
98	Farmer	Aruba
99	Haynes	Honduras
0		Chile
1		Djibouti
2		Antigua and Barbuda

```

86      Woods      Maldives
87      Vargas     Benin
88      Kane       Yemen
89      Flowers    Namibia
90      Chavez     Bhutan
91      Foley      Hungary
92      Ayers      Anguilla
93      Young      Solomon Islands
94      Douglas    Mali
95      Greer      Guyana
96      Atkinson  Sri Lanka
97      Guerra     Solomon Islands
98      Farmer     Aruba
99      Haynes     Honduras

```

```

0      Chile
1      Djibouti
2      Antigua and Barbuda
3      Dominican Republic
4      Slovakia (Slovak Republic)

```

...

```

95      Guyana
96      Sri Lanka
97      Solomon Islands
98      Aruba
99      Honduras

```

Name: Country, Length: 100, dtype: object

[]: